

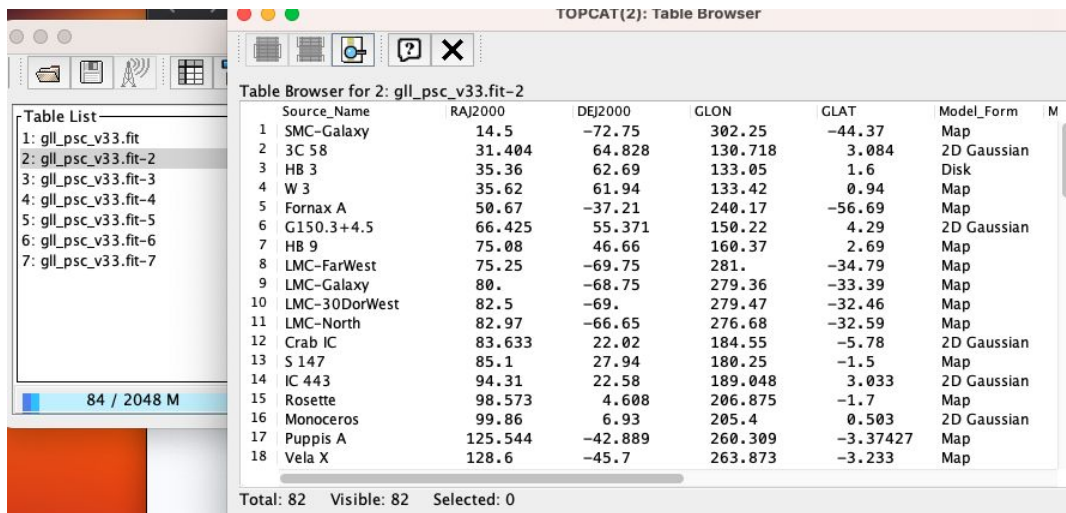
Applying ML/Deep Learning Techniques to Gamma Ray Astronomy

Krishiv Bhatia

Update: 12/08/2023

Update on Progress

- Refreshed Jerry's YouTube presentation
- Downloaded the 4FGL-DR4 FITS file from https://fermi.gsfc.nasa.gov/ssc/data/access/lat/14yr_catalog/
- Installed the TOPCAT tool on my laptop
- Viewed the 4FGL-DR4 FITS file from TOPCAT.



The screenshot shows the TOPCAT(2): Table Browser window. On the left, a 'Table List' sidebar shows several FITS files, with 'gll_psc_v33.fit-2' selected. The main panel displays a table titled 'Table Browser for 2: gll_psc_v33.fit-2'. The table has columns for Source_Name, RAJ2000, DEJ2000, GLON, GLAT, Model_Form, and M. It lists 18 sources, including SMC-Galaxy, 3C 58, HB 3, W 3, Fornax A, G150.3+4.5, HB 9, LMC-FarWest, LMC-Galaxy, LMC-30DorWest, LMC-North, Crab IC, S 147, IC 443, Rosette, Monoceros, Puppis A, and Vela X. At the bottom, it shows 'Total: 82', 'Visible: 82', and 'Selected: 0'.

	Source_Name	RAJ2000	DEJ2000	GLON	GLAT	Model_Form	M
1	SMC-Galaxy	14.5	-72.75	302.25	-44.37	Map	
2	3C 58	31.404	64.828	130.718	3.084	2D Gaussian	
3	HB 3	35.36	62.69	133.05	1.6	Disk	
4	W 3	35.62	61.94	133.42	0.94	Map	
5	Fornax A	50.67	-37.21	240.17	-56.69	Map	
6	G150.3+4.5	66.425	55.371	150.22	4.29	2D Gaussian	
7	HB 9	75.08	46.66	160.37	2.69	Map	
8	LMC-FarWest	75.25	-69.75	281.	-34.79	Map	
9	LMC-Galaxy	80.	-68.75	279.36	-33.39	Map	
10	LMC-30DorWest	82.5	-69.	279.47	-32.46	Map	
11	LMC-North	82.97	-66.65	276.68	-32.59	Map	
12	Crab IC	83.633	22.02	184.55	-5.78	2D Gaussian	
13	S 147	85.1	27.94	180.25	-1.5	Map	
14	IC 443	94.31	22.58	189.048	3.033	2D Gaussian	
15	Rosette	98.573	4.608	206.875	-1.7	Map	
16	Monoceros	99.86	6.93	205.4	0.503	2D Gaussian	
17	Puppis A	125.544	-42.889	260.309	-3.37427	Map	
18	Vela X	128.6	-45.7	263.873	-3.233	Map	

Update on Progress

- Downloaded ATNF Pulsar Catalogue tar gz file. Followed README to produce the executable psrcat file.
- Followed README
- Psrcat -h works but I get error in the db_file command
- What is this for? How to fix error?

```
rbhatia@Rajeshs-MacBook-Pro-Display psrcat_tar % ls
README          evalwrap.c      libpsrcat.a     psrcat.db
defineParams.c  evalwrap.o      makeit          psrcat.h
defineParams.o  externalCall.c  myrefs.bbl      psrcat_ref
displayOutput.c externalCall.o   parseParameters.c readCatalogue.c
displayOutput.o galcoord_ecliptic.c parseParameters.o readCatalogue.o
evaldefs.h      galcoord_ecliptic.o plotParams.c    wc_strncmp.c
evalkern.c      galcoord_ecliptic.o plotParams.o    wc_strncmp.h
evalkern.h      galcoord_ecliptic.o psrcat          wc_strncmp.o
evalkern.o      glitch.db       psrcat.c
rbhatia@Rajeshs-MacBook-Pro-Display psrcat_tar % psrcat -h
-all          Merge with $PSRCAT_RUNDIR/obs*.db
-allref       Lists the entire bibliographic reference file
-bib           Provide bibliographic reference
-bibliography  Lists references within the catalogue
-c            List of parameters to obtain from the database
-check        Checks the catalogue for incorrect entries
-db_file      Database path and name
-descend      Sort in descending order
-e            Produce short ephemeris files suitable for TEMPO
-e2           Produce long ephemeris files
-e3           Produce ephemeris file with selected parameters
-exactMatch   Require exact match for string comparisons
-h            This help information
-l            Logical string for filtering unwanted pulsars
              e.g. p0 > 5
-listref      Provide bibliographic references at end of table
-merge        Merges multiple catalogue files. For example
              -merge "mycat*.db" will merge the public catalogue
              with all the catalogues with filenames mycat*.db in the
              local directory.
-nohead       Do not print a header at the top of the table
-nonumber     Do not print line numbers in output
-nocand       Do not include pulsar candidates
-nointerim    Do not include interim solutions
-null        Null string for displaying for unknown parameter
```

```
rbhatia@Rajeshs-MacBook-Pro-Display psrcat_tar % psrcat -db_file ./psrcat.db -c "name p0 dm" -l "p0 > 3"
zsh: illegal hardware instruction psrcat -db_file ./psrcat.db -c "name p0 dm" -l "p0 > 3"
rbhatia@Rajeshs-MacBook-Pro-Display psrcat_tar %
rbhatia@Rajeshs-MacBook-Pro-Display psrcat_tar % psrcat -db_file ./psrcat.db
zsh: illegal hardware instruction psrcat -db_file ./psrcat.db
rbhatia@Rajeshs-MacBook-Pro-Display psrcat_tar %
```

Next Steps

- Could I have access to the Python scripts that you used with the 3FGL catalog
- Then, if I understand correctly, I need to reproduce your results using PyTorch?
- And then convert to using Neural Networks for Deep Learning?

Update: 01/08/2024

Progress

- 3FGLReproductionBasicLR Python code pointed to 3FGL Catalog file gll_psc_v16.fit.
- Updated it to point to 4FGL catalog file gll_psc_v33.fit.
- Tested with updated feature name
- Modified it for 1 hidden layer NN and DNN (2 hidden layers NN)

Results

Model	AGN Sensitivity	PSR Sensitivity	Accuracy of Best P Fit
Basic LR (0 hidden layer NN)	0.9724137931034482	1.0	0.9733333333333334
Neural Network (1 hidden layer)	0.9620689655172414	1.0	0.9633333333333334
Deep Neural Network (2 hidden layers)	0.9724137931034482	1.0	0.9733333333333334

Features Used

- Used the following features from columns in 4FGL catalog file gll_psc_v33.fit
 - ["PL_Index", "LP_Index", "PLEC_IndexS", "Variability_Index", "Unc_PL_Flux_Density", "Unc_LP_Flux_Density", "Unc_Energy_Flux100", "Unc_PLEC_Flux_Density", "Unc_Flux1000", "LP_beta", "Frac_Variability", "LP_SigCurv", "Signif_Avg"]
 - Found the corresponding 3FGL columns in the 4FGL list
 - Did some trial and error for other columns and checked model performance
- Column list is in pages 32-33 of Abdollahi_2020_ApJS_247_33 paper

Columns

- Printed out from code

1 DataRelease	21 PL_Flux_Density	41 Unc_PLEC_Exp_Index	61 ASSOC_4FGL
2 RAJ2000	22 Unc_PL_Flux_Density	42 PLEC_SigCurv	62 TEVCAT_FLAG
3 DEJ2000	23 PL_Index	43 PLEC_EPeak	63 ASSOC_TEV
4 GLON	24 Unc_PL_Index	44 Unc_PLEC_EPeak	64 CLASS1
5 GLAT	25 LP_Flux_Density	45 Npred	65 ASSOC1
6 Conf_68_SemiMajor	26 Unc_LP_Flux_Density	46 Flux_Band	66 ASSOC_PROB_BAY
7 Conf_68_SemiMinor	27 LP_Index	47 Unc_Flux_Band	67 ASSOC_PROB_LR
8 Conf_68_PosAng	28 Unc_LP_Index	48 nuFnu_Band	68 RA_Counterpart
9 Conf_95_SemiMajor	29 LP_beta	49 Sqrt_TS_Band	69 DEC_Counterpart
10 Conf_95_SemiMinor	30 Unc_LP_beta	50 Variability_Index	70 Unc_Counterpart
11 Conf_95_PosAng	31 LP_SigCurv	51 Frac_Variability	71 Flags
12 ROI_num	32 LP_EPeak	52 Unc_Frac_Variability	72 ASSOC_FGL
13	33 Unc_LP_EPeak	53 Signif_Peak	73 ASSOC_FHL
Extended_Source_Name	34 PLEC_Flux_Density	54 Flux_Peak	74 ASSOC_GAM1
14 Signif_Avg	35 Unc_PLEC_Flux_Density	55 Unc_Flux_Peak	75 ASSOC_GAM2
15 Pivot_Energy	36 PLEC_IndexS	56 Time_Peak	76 ASSOC_GAM3
16 Flux1000	37 Unc_PLEC_IndexS	57 Peak_Interval	77 CLASS2
17 Unc_Flux1000	38 PLEC_ExpfactorS	58 Flux_History	78 ASSOC2
18 Energy_Flux100	39 Unc_PLEC_ExpfactorS	59 Unc_Flux_History	
19 Unc_Energy_Flux100	40 PLEC_Exp_Index	60 Sqrt_TS_History	
20 SpectrumType			

Feature Transformation

- FGL4Dataset feature transformations

```
def __init__(self, wanted_type_col, fluxindices, wantedtypes, wantedfeaturenames, pointsourcecata):  
    # Create x/input data  
    xdata = []  
    ydata = []  
    # Select sources that are a part of the wanted types  
    for source in pointsourcecatalogue.data:  
        for feature_names in wantedfeaturenames:  
            print("{0} = {1}".format(*args: feature_names, source[feature_names]))  
            if source[wanted_type_col] in wantedtypes:  
                unit = []  
                # Append wanted params. Applied a log transformation to paras with highly skewed distributions.  
                unit.append(float(source[wantedfeaturenames[0]]))  
                unit.append(float(source[wantedfeaturenames[1]]))  
                unit.append(float(source[wantedfeaturenames[2]]))  
                unit.append(float(source[wantedfeaturenames[3]]))  
                unit.append(math.log(source[wantedfeaturenames[4]]))  
                if (source[wantedfeaturenames[4]] > 0) else float(source[wantedfeaturenames[4]])  
                unit.append(math.log(source[wantedfeaturenames[5]]))  
                if (source[wantedfeaturenames[5]] > 0) else float(source[wantedfeaturenames[5]])  
                unit.append(math.log(source[wantedfeaturenames[6]]))  
                if (source[wantedfeaturenames[6]] > 0) else float(source[wantedfeaturenames[6]])  
                unit.append(math.log(source[wantedfeaturenames[7]]))  
                if (source[wantedfeaturenames[7]] > 0) else float(source[wantedfeaturenames[7]])  
                unit.append(math.log(source[wantedfeaturenames[8]]))  
                if (source[wantedfeaturenames[8]] > 0) else float(source[wantedfeaturenames[8]])  
                unit.append(float(source[wantedfeaturenames[9]]))  
                unit.append(float(source[wantedfeaturenames[10]]))  
                unit.append(float(source[wantedfeaturenames[11]]))  
                unit.append(float(source[wantedfeaturenames[12]]))
```

Apply log transformations
to params with highly
skewed distributions

PL_Index = 2.5443360805511475

LP_Index = 2.5455660820007324

PLEC_IndexS = 2.539679765701294

Variability_Index = 25.699934005737305

Unc_PL_Flux_Density = 3.7675866251677195e-14

Unc_LP_Flux_Density = 4.284851688957439e-14

Unc_Energy_Flux100 = 4.238916617889388e-13

Unc_PLEC_Flux_Density = 3.952874743566419e-14

Unc_Flux1000 = 2.748007377206818e-11

LP_beta = -0.009841139428317547

Frac_Variability = 0.5357248187065125

LP_SigCurv = 0.0

Signif_Avg = 8.640633583068848

Neural Networks Configurations Used

- Used

- 0 hidden layer Neural Networks - Linear Regression
- 1 hidden layer NN
- Deep Neural Networks (2 hidden layers)

0 hidden layer NN - Linear Regression

```
Linear(in_features=13, out_features=1, bias=True)
```

NN (1 hidden layer)

```
Sequential(  
  (0): Linear(in_features=13, out_features=60, bias=True)  
  (1): ReLU()  
  (2): Linear(in_features=60, out_features=1, bias=True)  
)
```

DNN (2 hidden layers)

```
Sequential(  
  (0): Linear(in_features=13, out_features=250, bias=True)  
  (1): ReLU()  
  (2): Linear(in_features=250, out_features=40, bias=True)  
  (3): ReLU()  
  (4): Linear(in_features=40, out_features=1, bias=True)  
)
```

Other Changes & Github

Classes

- Separated the large code into classes for code modularity, improvement
- Created separate classes for:
 - **Driver:** 3FGLReproductionBasicLR_v1.py, 4FGLReproductionBasic{LR, NN, DNN}.py,
 - **Feature transformation:** FGL3Dataset.py / FGL4Dataset.py
 - **Models:** LogisticRegression.py, NeuralNetwork.py, DeepNeuralNetwork.py
 - **Utility functions:** utils.py (e.g. run_iteration)

- 3FGLReproductionBasicLR_v1.py
 - FGL3Dataset.py
 - LogisticRegression.py
 - utils.py

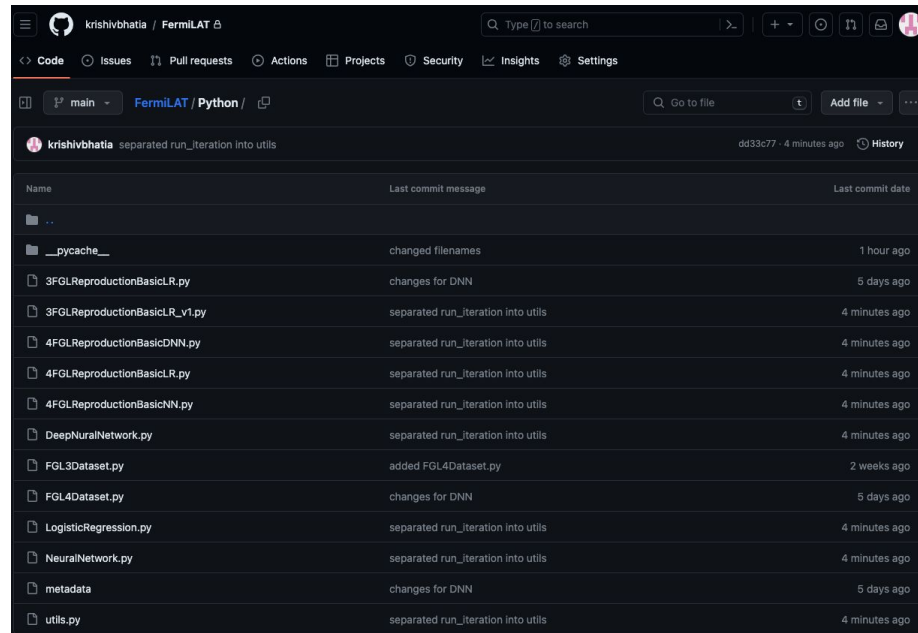
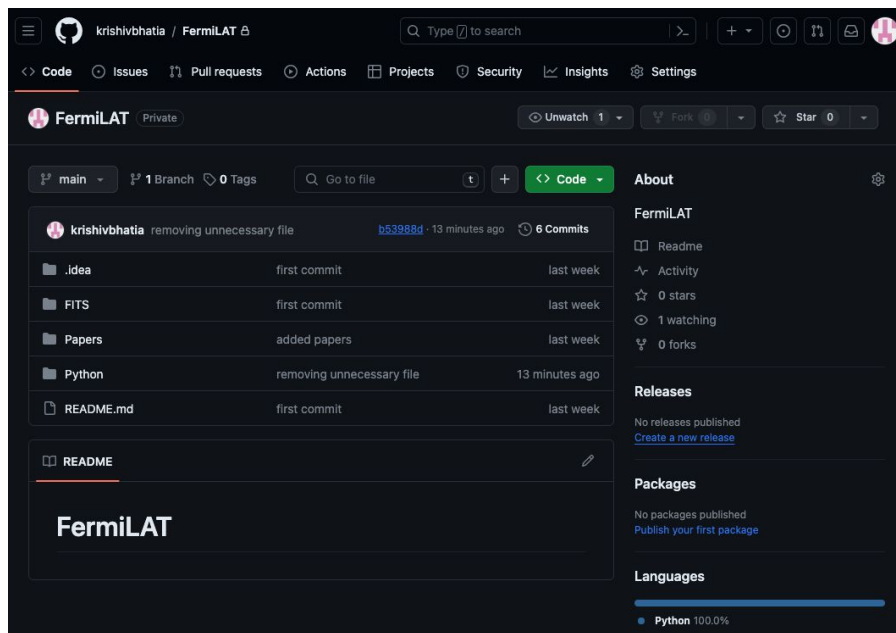
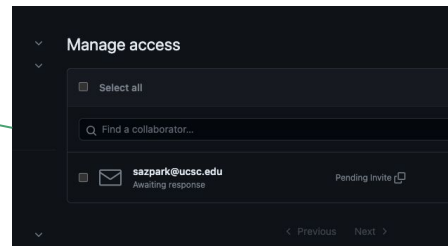
- 4FGLReproductionBasicLR.py
 - FGL4Dataset.py
 - LogisticRegression.py
 - utils.py

- 4FGLReproductionBasicLR.py
 - FGL4Dataset.py
 - LogisticRegression.py
 - utils.py
- 4FGLReproductionBasicDNN.py
 - FGL4Dataset.py
 - DeepNeuralNetwork.py
 - utils.py

Github Repo

- Checked into github
 - <https://github.com/krishivbhatia/FermiLAT/>

Sent you collaborator invite



Results: Linear Regression

Files for Linear Regression

- Python Scripts
 - 4FGLReproductionBasicLR.py
 - FGL4Dataset.py
 - LogisticRegression.py
 - utils.py
- Execution
 - python3 4FGLReproductionBasicLR.py

1-layer NN (No hidden layer)
Basic LR



```
main ▾ FermiLAT / Python / LogisticRegression.py   
vrbhatia changes for DNN  
1 loc) · 1.79 KB  
Blame      
  
import math  
import torch  
import torch.nn as nn  
  
class LogisticRegression(nn.Module):  
    def __init__(self, input_features):  
        print("(len(input_features[0][0]) = ", len(input_features[0][0]))  
        print("input_features[0][0] = ", input_features[0][0])  
        super(LogisticRegression, self).__init__()  
        self.model = nn.Linear(len(input_features[0][0]), 1)  
        print()  
        print(self.model)  
        print()  
  
    def forward(self, x):  
        y_predicted = torch.sigmoid(self.model(x))  
        return y_predicted
```

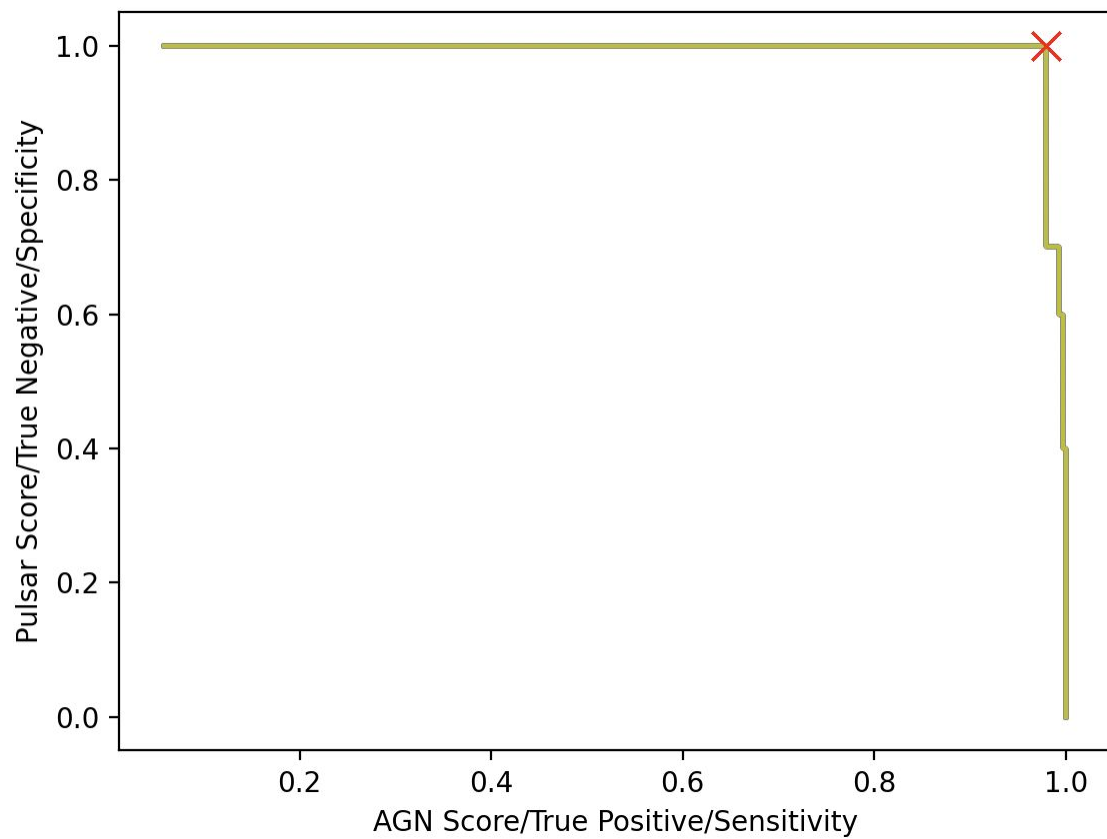
Results

- Linear(in_features=13, out_features=1, bias=True)
- **Results (200 iterations, lr = 0.00001)**
 - optimal_p = 0.9352
 - Best True Positive (AGN Successfully Identified): 282 (290)
 - Best AGN/True Positive/Sensitivity : 0.9724137931034482
 - Best True Negative (PSR Successfully Identified): 10 (10)
 - Best Pulsar/True Negative/Specificity : 1.0
 - Accuracy of Best P Fit : 0.9733333333333334

Results

- Linear(in_features=13, out_features=1, bias=True)
- **Results (300 iterations, lr = 0.00001)**
 - optimal_p = 0.9352
 - Best True Positive (AGN Successfully Identified): 282 (290)
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 - Best True Negative (PSR Successfully Identified): 10 (10)
 - Best Pulsar/True Negative/Specificity : 1.0
 - Accuracy of Best P Fit : 0.9733333333333334

Plot



Results: NN - 1 hidden layer

Files for NN (1 hidden layer)

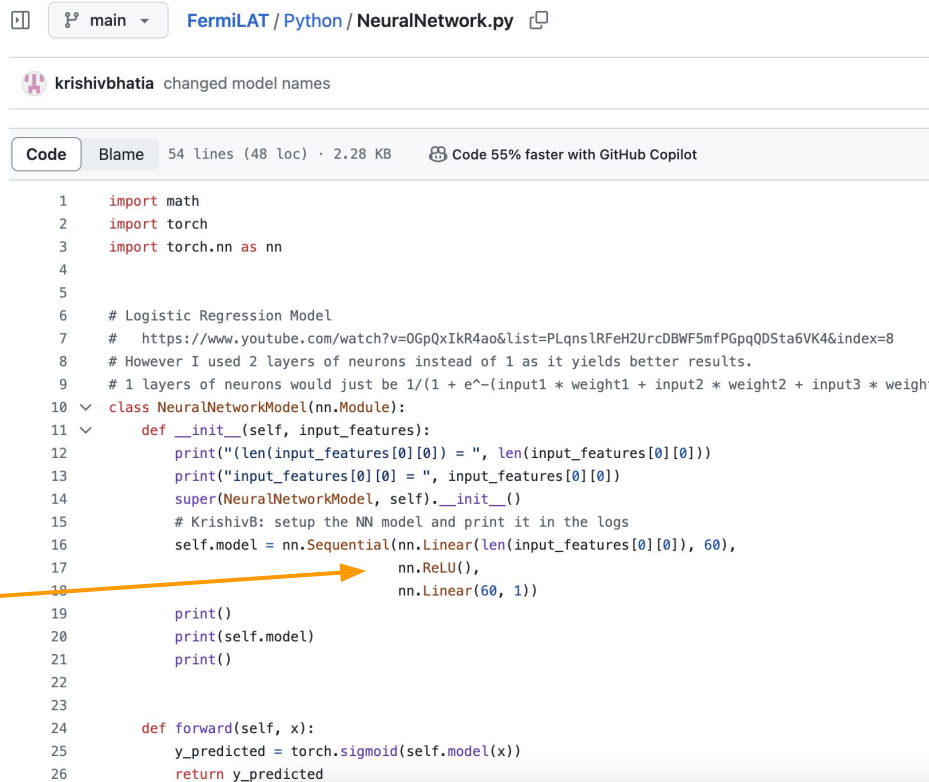
- Python Scripts

- 4FGLReproductionBasicNN.py
- FGL4Dataset.py
- NeuralNetwork.py
- utils.py

- Execution

- python3 4FGLReproductionBasicNN.py

1 hidden layers NN
ReLU activation function



```
main FermiLAT / Python / NeuralNetwork.py
krishivbhatia changed model names

Code Blame 54 lines (48 loc) · 2.28 KB Code 55% faster with GitHub Copilot

1 import math
2 import torch
3 import torch.nn as nn
4
5
6 # Logistic Regression Model
7 # https://www.youtube.com/watch?v=0GpQxIkR4ao&list=PLqns1RFeh2UrcDBWF5mfPGpqQDSta6VK4&index=8
8 # However I used 2 layers of neurons instead of 1 as it yields better results.
9 # 1 layers of neurons would just be 1/(1 + e^~(input1 * weight1 + input2 * weight2 + input3 * weight3))
10 class NeuralNetworkModel(nn.Module):
11     def __init__(self, input_features):
12         print("(len(input_features[0][0]) = ", len(input_features[0][0]))
13         print("input_features[0][0] = ", input_features[0][0])
14         super(NeuralNetworkModel, self).__init__()
15         # KrishivB: setup the NN model and print it in the logs
16         self.model = nn.Sequential(nn.Linear(len(input_features[0][0]), 60),
17                                   nn.ReLU(),
18                                   nn.Linear(60, 1))
19
20         print()
21         print(self.model)
22         print()
23
24     def forward(self, x):
25         y_predicted = torch.sigmoid(self.model(x))
26         return y_predicted
```

Results

- **Model: Sequential(**

- (0): Linear(in_features=13, out_features=60, bias=True)

- (1): ReLU() ← Activation Function

- (2): Linear(in_features=60, out_features=1, bias=True))

1 hidden layer



- **Results (300 iterations, lr = 0.0001)**

- optimal_p = 0.9525
 - Best True Positive (AGN Successfully Identified): 279 (290)
 - Best AGN/True Positive/Sensitivity : 0.9620689655172414
 - Best True Negative (PSR Successfully Identified): 10 (10)
 - Best Pulsar/True Negative/Specificity : 1.0
 - Accuracy of Best P Fit : 0.9633333333333334

Results

- **Model: Sequential**
 - (0): Linear(in_features=13, out_features=60, bias=True)
 - (1): ReLU()
 - (2): Linear(in_features=60, out_features=1, bias=True))
- **Results (60 iterations, lr = 0.0001)**
 - optimal_p = 0.9525
 - Best True Positive (AGN Successfully Identified): 279 (290)
 - Best AGN/True Positive/Sensitivity : 0.9620689655172414
 - Best True Negative (PSR Successfully Identified): 10 (10)
 - Best Pulsar/True Negative/Specificity : 1.0
 - Accuracy of Best P Fit : 0.9633333333333334

Results

- **Model: Sequential**
 - (0): Linear(in_features=13, out_features=40, bias=True)
 - (1): Tanh() ← tested with tanh() activation function also
 - (2): Linear(in_features=40, out_features=1, bias=True))
- **Results (300 iterations, lr = 0.0001)**
 - optimal_p = 0.9544
 - Best True Positive (AGN Successfully Identified): 278 (290)
 - Best AGN/True Positive/Sensitivity : 0.9586206896551724
 - Best True Negative (PSR Successfully Identified): 10 (10)
 - Best Pulsar/True Negative/Specificity : 1.0
 - Accuracy of Best P Fit : 0.96

Results: DNN - 2 hidden layers

Files for DNN: 2 hidden layers

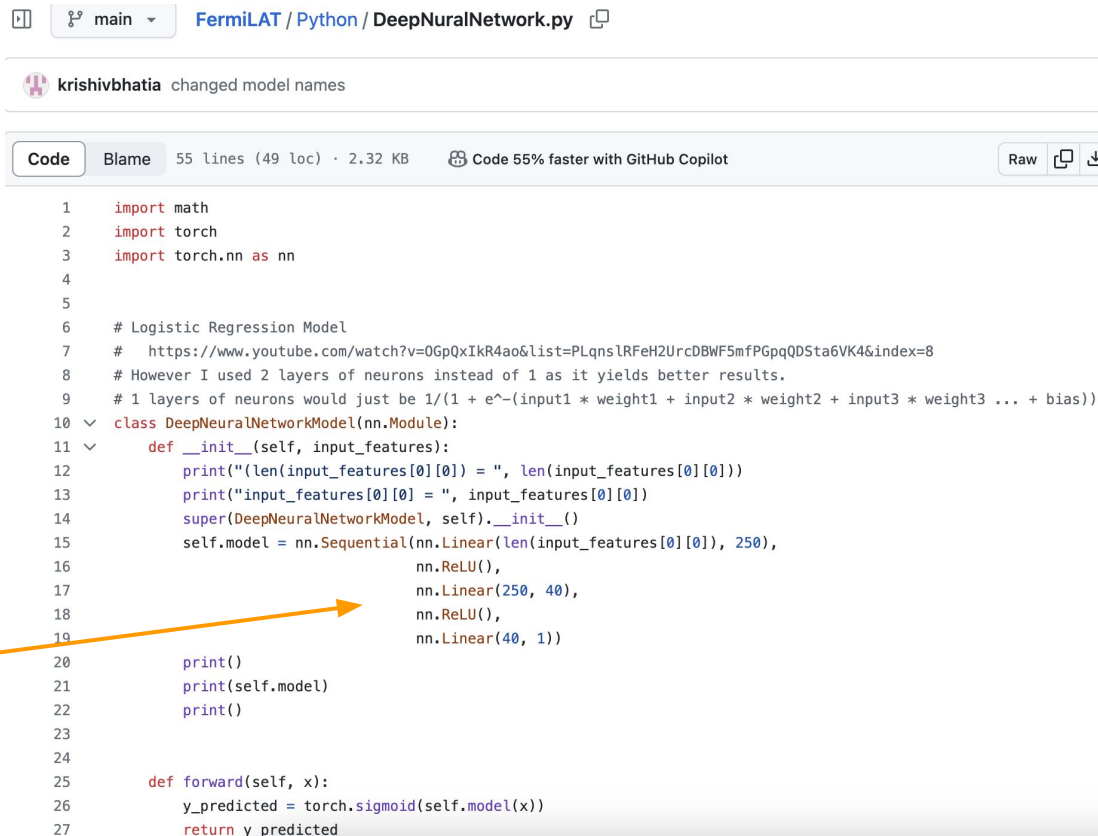
- Python Scripts

- 4FGLReproductionBasicDNN.py
- FGL4Dataset.py
- DeepNeuralNetwork.py
- utils.py

- Execution

- python3 4FGLReproductionBasicDNN.py

2 hidden layers NN
ReLU activation function



```
1 import math
2 import torch
3 import torch.nn as nn
4
5
6 # Logistic Regression Model
7 # https://www.youtube.com/watch?v=OGpQxIkR4ao&list=PLqnsIRFeH2UrcDBWF5mfPGpqQDSta6VK4&index=8
8 # However I used 2 layers of neurons instead of 1 as it yields better results.
9 # 1 layers of neurons would just be 1/(1 + e^-(input1 * weight1 + input2 * weight2 + input3 * weight3 ... + bias))
10 class DeepNeuralNetworkModel(nn.Module):
11     def __init__(self, input_features):
12         print("len(input_features[0][0]) = ", len(input_features[0][0]))
13         print("input_features[0][0] = ", input_features[0][0])
14         super(DeepNeuralNetworkModel, self).__init__()
15         self.model = nn.Sequential(nn.Linear(len(input_features[0][0]), 250),
16                                   nn.ReLU(),
17                                   nn.Linear(250, 40),
18                                   nn.ReLU(),
19                                   nn.Linear(40, 1))
20
21         print()
22         print(self.model)
23         print()
24
25     def forward(self, x):
26         y_predicted = torch.sigmoid(self.model(x))
27         return y_predicted
```

Results

- **Model: Sequential(**

- (0): Linear(in_features=13, out_features=250, bias=True)

- (1): ReLU() ← activation function

- (2): Linear(in_features=250, out_features=40, bias=True)

- (3): ReLU() ← activation function

- (4): Linear(in_features=40, out_features=1, bias=True))

2 hidden layers



- **Results (200 iterations, lr = 0.0001)**

- optimal_p = 0.9475
 - Best True Positive (AGN Successfully Identified): 282 (290)
 - Best AGN/True Positive/Sensitivity : 0.9724137931034482
 - Best True Negative (PSR Successfully Identified): 10 (10)
 - Best Pulsar/True Negative/Specificity : 1.0
 - Accuracy of Best P Fit : 0.9733333333333334

Results

- **Model: Sequential()**
 - (0): Linear(in_features=13, out_features=250, bias=True)
 - (1): ReLU()
 - (2): Linear(in_features=250, out_features=40, bias=True)
 - (3): ReLU()
 - (4): Linear(in_features=40, out_features=1, bias=True))
- **Results (300 iterations, lr = 0.0001)**
 - optimal_p = 0.9475
 - Best True Positive (AGN Successfully Identified): 282 (290)
 - Best AGN/True Positive/Sensitivity : 0.9724137931034482
 - Best True Negative (PSR Successfully Identified): 10 (10)
 - Best Pulsar/True Negative/Specificity : 1.0
 - Accuracy of Best P Fit : 0.9733333333333334

Questions

- Hardness ratios of LAT energy bands
 - 3FGL Catalog had many bands:
 - "Flux100_300", "Flux300_1000", "Flux1000_3000", "Flux3000_10000", "Flux10000_100000"
 - 4FGL Catalog has only one energy band column present:
 - Energy_Flux100
 - From Abdollahi 2020, definition: *Energy flux from 100MeV to 100GeV obtained by spectral fitting*
 - Where do we get other bands to calculate hardness ratios?

Next Steps

- Random Forest & Decision Tree implementations
 - Found Random Forest & Decision Tree implementations using PyTorch
 - <https://discuss.pytorch.org/t/how-can-i-use-knn-random-forest-models-in-pytorch/42464/2>
 - https://github.com/ValentinFigue/Sklearn_PyTorch
 - Test class shows how to use it:
 - https://github.com/ValentinFigue/Sklearn_PyTorch/blob/master/tests/random_forest_tests.py
 - https://github.com/ValentinFigue/Sklearn_PyTorch/blob/master/tests/binary_tree_tests.py
 - Use it to implement Random Forest & Decision Trees in PyTorch for 4FGL Catalog?